Diagnosis-Guided Compressed Latent Replay for Continual Document Understanding

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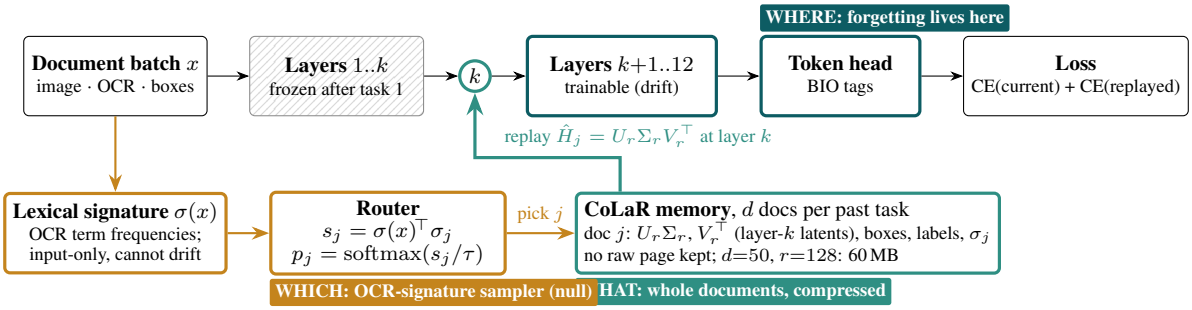


Figure 1. **Diagnosis-guided compressed latent replay (CoLaR).** The anatomy of Sec. 4 fixes *where* to act (head and late layers) and *what* memory must hold (whole documents, Sec. 7): each past document is stored as a truncated SVD of its layer-k activations (60 MB per d=50 task) and rebuilt during replay. A drift-free OCR-signature sampler probes *which* stored document to replay and is reported as a three-seed null (Sec. 7).

Abstract

Continual learning for document understanding is dominated by raw-exemplar replay, which deployed pipelines cannot keep for privacy reasons. We take a diagnosis-first route. Fisher-weighted displacement and layer-wise CKA across four backbones (LayoutLMv3, LiLT, BROS, BERT), three scenarios and three seeds show that forgetting in continual key-information extraction is output-localised*, at the classifier head and late layers, and* architecture-general*. Protecting the locus does not remove forgetting but* relocates *it, so a buffer-free fix cannot work: five such families fail for the same readout-marginal snap. Guided by the diagnosis we propose* ***CoLaR****, per-document truncated-SVD latent replay that preserves the (feature, position, label) binding every marginal summary destroys (+27 AA single-variable control):* 87.8±0.5 *AA on DIL at* 60 *MB, within* 1.1 *of the joint oracle and matching raw replay at* 2.7× *less storage, with no raw bytes stored. Under class-incremental label growth the shared softmax, not the memory, is the bottleneck: scored on each session's own label set, the oracle rises from* 33 *to* 92 *AA while every replay row stays near* 20–30*. Giving each session its own background logit (****CoLaR-SBR****, routed per token with no task id) reaches* 88.0±1.9 *on CIL-CORD and* 89.7 *on the mixed stream, at the oracle; reading each session's block under a stored per-session LoRA delta (****SBR-Recall****) reaches* 92.8±0.5 *with exact retention. On image CIL the same bank is beaten by a Gaussian-head method whose document analogue is null, isolating whole-document binding as the modality-discriminating ingredient.*

1. Introduction

Organisations process a continuous stream of visually-rich documents (forms, invoices, receipts, contracts) whose meaning depends not only on the words but on where those words sit on the page. Key-information extraction (KIE) from such documents is cast as token-level sequence labelling over a tri-modal input (OCR text, 2-D layout, page image), and multimodal document encoders such as LayoutLMv3 [14], LiLT [37] and BROS [12] define the state of the art. Deployed systems are not static: new document types appear, schemas grow and the input drifts, while retraining on all history is blocked by cost and by privacy constraints on earlier data. This is continual learning (CL), whose central obstacle is catastrophic forgetting [25].

The CL literature offers a mature toolbox: importance-weighted regularisation [15], distillation from past models [20], exemplar replay [3, 32], task prompts [34, 40, 41] and per-task low-rank adapters [39], but almost all of it was developed on unimodal image or text classification, and it treats the network as one homogeneous block: a single quadratic penalty across all parameters, whole-network output matching, or a depth-wise freezing cut chosen by hand. A tri-modal document encoder is not homogeneous: it has a text stream, a visual stream, layout embeddings, attention blocks at increasing depth and a task head. Parameter-efficient and modular families sharpen rather than resolve this, because deciding *which* weights to adapt or *where* to inject a prompt presupposes knowing which part of the encoder is at risk.

Problem and approach. Under continual token-level KIE (a domain-incremental sequence over FUNSD, SROIE and CORD, a class-incremental split of CORD, and a mixed sequence that alternates the two), privacy constraints on earlier documents rule out raw-exemplar replay. The literature's buffer-free alternatives perform below replay by a large margin, and no existing analysis says why in a document setting. We follow a diagnosis-first route. First we localise forgetting inside a document encoder (Sec. 4), then show that the diagnosis explains the remedy ordering, including a backbone-dependent sign flip in parameter-space remedies (Sec. 5). The diagnosis makes a sharp prediction: because the head is where forgetting lives *and* the drift relocates when the head is protected, a buffer-free fix cannot work; we falsify five consolidation families against this prediction (Sec. 6). Finally we read off what a buffer *must* contain, and build it (Sec. 7): per-document SVD-compressed latent replay (**CoLaR**) preserves the (feature, position, label) binding of a document while compressing it, matches raw-exemplar replay at $2.7\times$ less storage, and never leaves raw document bytes in the memory. Where the label space grows, the same diagnosis points at the readout rather than the memory, and a session-blocked readout (**CoLaR-SBR**) closes the class-incremental gap to the oracle (Sec. 8).

Contributions.

1. An *anatomy* of forgetting in document IE over four backbones and three scenarios: output-localised, architecture-general, and conserved under constraint, since protecting the diagnosed locus relocates the drift to whatever stays plastic (Sec. 4 and Fig. 2).
2. A prediction-tested benchmark of six classical strategies ($68/72$ cells, three seeds) whose ordering the diagnosis explains, including a 17-point sign flip in EWC's benefit across backbones, and a convergent negative over five buffer-free families with one mechanistic cause, a readout-marginal snap, closed by registered kill-tests (Secs. 5 and 6).
3. **CoLaR**: per-document truncated-SVD latent replay that stores no raw bytes, reaches 87.8 ± 0.5 AA on DIL at 60 MB (within 1.1 of the joint oracle, matching raw-exemplar ER at $2.7\times$ less storage), and isolates whole-document binding as the necessary buffer ingredient ($+27$ AA); on image CIL the same bank is beaten by a Gaussian head whose document analogue is null (Sec. 7).
4. **Label growth is a readout problem.** The CIL-CORD "floor" is largely a scoring artefact (oracle 33 vs. 92 under session-restricted scoring); what remains is a contradiction on the single background logit. **CoLaR-SBR** gives each session its own block $[\mathrm{O}_b, \mathrm{tags}_b]$ and routes per token with no task id: 88.0 ± 1.9 on CIL-CORD and 89.7 on Mixed, at the oracle. **SBR-Recall** reads each block under a stored per-session LoRA delta of the upper trunk: 92.8 ± 0.5 over three seeds, above the oracle, with flat retention (Sec. 8 and Tab. 3).

2. Related Work

Continual-learning families. Following Wang et al. [38], remedies fall into regularisation (EWC [15], SI [44], MAS [1]; output-side distillation, LwF [20]), replay (GEM [22], ER [32], DER++ [3], iCaRL [31]), optimisation-geometry (GPM [33], OGD [6], C-Flat [2]), representation-based methods over a frozen backbone (L2P [41], DualPrompt [40], CODA-Prompt [34], O-LoRA [39], CL-LoRA [11], RanPAC [26]), and architecture-based isolation [23, 43]. Replay is empirically the strongest family [24, 38] but stores raw data, which document deployments cannot keep. The buffer-free alternatives we test in Sec. 6 are drawn from the strongest current members of each line: model merging with classifier realignment (LCA [35]), subspace transfer, and exemplar-free *feature* replay, class-conditional Gaussians in a frozen feature space (PASS, FeTrIL, FeCAM [8, 29, 52]).

Latent and compressed replay. CoLaR descends from methods that store mid-network activations instead of raw inputs: REMIND [10] quantises later-layer features, Latent Replay [28] banks activations at a chosen layer, and brain-inspired generative replay [36] samples a per-class VAE. Each stores one summary per token and targets class labels; CoLaR stores one SVD per document for token-level BIO, so it must keep the binding between a token's feature, position and label, which Sec. 7 shows is exactly what every marginal summary destroys.

Where forgetting lives. Ramasesh et al. [30] localise forgetting in split image classification to the layers nearest the output, and Zhai et al. [45] find the same in multimodal LLMs used as classifiers. Multimodal CL benchmarks hint

that architecture shapes forgetting (dual-stream models forget less, deeper layers more [48]), and per-layer budget routing (D-MoLE [7]) assumes non-uniform need without measuring it. None attributes forgetting to the components of a tri-modal encoder, tests whether the location is architecture-invariant, or asks what the drift does when the located component is protected; we do all three.

Continual information extraction. Continual NER handles the semantic shift of "O" with distillation and pseudo-labelling (CPFD [46], CFNER [50], ICE [21], SpanKL [51]) or span generation (GenCNER [42]); continual RE pairs episodic memory with prototypes (EMAR [9], RP-CRE [5], CRL [49]) or prompt pools (WAVE-CRE [18]). All are sentence-level and text-only. Document-level CL is nascent: GDI-Bench [19] mitigates MLLM forgetting by heuristic layer freezing, continual table detection [27] uses plain ER, and ProtoNER [17] reaches LayoutLM-family encoders only few-shot. No prior work diagnoses which component of a document encoder forgets or evaluates the classical CL families on multimodal KIE across backbones under one protocol.

3. Setup

Task. A document is a triple $D = (W, B, I)$ of OCR tokens, normalised bounding boxes and the page image; the model f_θ emits BIO labels per token and is trained with token-level cross-entropy. Performance is entity-level F1 with BIO-aware span matching (`seqeval`). In the continual setting the learner sees tasks $\mathcal{T}_1, \ldots, \mathcal{T}_T$ and at task t may access only $\mathcal{D}_t$ and, for replay methods, a bounded memory. OCR and boxes are dataset-provided throughout, so no OCR engine enters the loop.

Datasets and scenarios. We use FUNSD (149/50 train/test forms, 7 BIO tags), CORD (800/100/100 receipts, 30 fine-grained classes, 61 tags) and SROIE (626/347 receipts, 9 tags). **DIL**: FUNSD→SROIE→CORD under a fixed unified schema {HEADER, KEY, VALUE, OTHER}; the label space is constant and only the input distribution shifts, so forgetting is pure representation drift. **CIL-CORD**: CORD's 30 classes introduced six at a time over five sessions in a fixed easy-to-hard curriculum; the head grows, and tokens of not-yet-introduced classes are absorbed into "O", whose meaning therefore shifts. **Mixed**: a six-session sequence alternating new-class transitions within CORD with new-domain transitions to FUNSD and SROIE, ending in a FUNSD revisit. Session details are in the supplement (Sec. A).

Backbones. LayoutLMv3-base [14] is the primary tri-modal encoder (text, layout, image; single-stream). LiLT-base [37] and BROS-base [12] are layout-aware but vision-free; BERT-base is a text-only external comparator. Every method runs unchanged on all four through one wrapper contract; heads are plain linear layers.

Methods. Every comparison uses one frozen reference set of twelve rows: two bounds, naive sequential fine-tuning (lower) and joint multi-task training on the pooled data (oracle upper), and ten published baselines, one per family and era. They are ER [32] and DER++ [3] for replay (buffer 200 documents or 2,000 images; DER++ $\alpha{=}\beta{=}0.5$), LwF [20] for distillation ($T{=}2$), CODA-Prompt [34] for prompts, CL-LoRA [11] for parameter-efficient adaptation (rank 8), LCA [35] for model merging, and two *modality-paired* slots holding the strongest head-side and representation-side method of each modality: SLCA [47] and RanPAC [26] on image class-incremental learning, CPFD [46] and IS3 [4] on document continual learning. Dominated relatives (EWC, L2P, DualPrompt, O-LoRA, ER+C-Flat++) appear only in the generality grid (EWC, Tab. 1) or the supplement (Sec. B). The falsified families of §6 and the replay controls of §7 run on LayoutLMv3/DIL unless stated.

Protocol. AdamW, learning rate $5{\times}10^{-5}$, weight decay 0.01, gradient clipping 1.0. Each task trains *to convergence*: early stopping on the task's validation entity-F1 (patience 2, $\delta{=}0.1$) under a 100-epoch ceiling that never binds, restoring the best-validation checkpoint at each boundary. FUNSD and SROIE have no validation split; for them the stopping signal is read from the evaluation set, a mild optimistic bias shared by every method. Grid results are mean ± s.d. over seeds $\{42, 7, 123\}$. *Session-restricted scoring.* In CIL-CORD and the CIL halves of Mixed, session j's test gold hides every other session's entity as "O", so a model that correctly labels another session's entity on the same receipt is charged a false positive. Tabs. 1 and 2 use this legacy scoring; Tab. 3 restricts predictions to session j's label set when computing $R[t, j]$ (a no-op on DIL), which is the scoring the CIL literature's per-task matrices assume.

Metrics and the localisation protocol. From $R[i, j]$ (F1 on task j after training task i) we report AA $= \frac{1}{T}\sum_j R[T, j]$ and BWT $= \frac{1}{T-1}\sum_{j<T}(R[T, j]{-}R[j, j])$; forward transfer is uninformative under label growth. Localisation is one backbone-agnostic protocol over a checkpoint pair $(\theta^{(t-1)}, \theta^{(t)})$, a partition of the parameters into groups g (by component or by depth) and an old-task probe set. *Localise*: per-group old-task-Fisher-weighted displacement $D_g = \sum_{\phi\in g} F_\phi^{(t-1)}(\theta_\phi^{(t)} - \theta_\phi^{(t-1)})^2$, large exactly where parameters that mattered for the previous task were overwritten, and per-token linear CKA [16] between the checkpoints at one probe per group. *Test*: dominant group vs. each rival (Mann–Whitney over seeds, Bonferroni) and, across back-

bones, a permutation test on the normalised D_g profile for a shared location. *Perturb*: protect the located group, retrain, re-run. D_g is distinct from Fisher importance F_g, which peaks at the head for any token classifier by construction and is never read as forgetting.

4. Anatomy of Forgetting

Forgetting is total within one boundary, on every backbone. Under naive sequential fine-tuning the sub-diagonal of R collapses to 1–3 F1 (LayoutLMv3 loses FUNSD from 89.1 to 3.3 after one task; LiLT 83.9→2.1, BROS 88.9→6.6). The DIL naive floor is 41.3/40.1/40.3/39.4 on LayoutLMv3/LiLT/BROS/BERT against joint 88.7/83.8/87.5/77.2, and on Mixed the joint-to-naive gap agrees to 2.5 points across backbones (Tab. 1).

Finding 1: output-localised. Two instruments agree. Per-token CKA between consecutive checkpoints falls monotonically with depth: 1.00 at the text/layout/patch embeddings, 0.78 over early encoder layers, 0.34 mid, 0.25 at the last layer and 0.16 at the classifier. Old-task-Fisher-weighted displacement D_g (supplementary Sec. C) is one to four orders of magnitude larger at the head than at any encoder bucket, on all four modality-mask conditions of LayoutLMv3, on BERT, LiLT and BROS. A per-component test rejects the null of uniform displacement with the head dominant ($p < 0.01$). The locus is not fusion-specific: the full model, text+layout and text-only LayoutLMv3 forget comparably (BWT −81, −88, −79), so forgetting is carried by the text-and-head path common to all three; with text removed, two of three seeds collapse to an all-background predictor.

Finding 1′: architecture-general. Permuting LiLT's and BROS's per-depth displacement profiles against the LayoutLMv3 reference fails to reject a shared *location* ($p = 0.38, 0.51$), and BERT reproduces the same head-dominant, depth-monotone pattern. We claim only shared location; cross-backbone *magnitude* comparisons are underpowered at three seeds.

Finding 2: protecting the locus relocates the drift. If forgetting lives at the head and late layers, the natural intervention is to protect them. On DIL/LayoutLMv3 (seed 42) freezing everything after task 1 gives AA 41.1, BWT −0.0: nothing is forgotten and nothing is learned. Freezing only the diagnosed head and late layers, leaving the early and mid encoder plastic, gives AA 38.8 with BWT −76.8, naive-level forgetting, while early/mid CKA, which is ≈ 1.0 under naive training, collapses to 0.19. The drift did not disappear; it migrated to whatever remained plastic. A control that adds a memory to the frozen locus without protecting it (38.9, −75.6) is indistinguishable, so drift, not the absence of a memory, is the bottleneck. Uniform protection fails for the same reason: a consolidation penalty spread equally over every bucket collapses to 42.6 AA, whereas the same budget concentrated on the head/late locus recovers 84.7, and head-only (86.1) or late-only (86.2) each recover essentially all of it (three seeds). Under architectural constraint the total forgetting is redistributed, not removed; Fig. 2 extends the freeze arm to three task orders and a long class-incremental stream.

Prediction. The anatomy is output-facing and conserved under constraint. It therefore predicts that (i) only a mechanism that re-grounds the head on old-task targets can recover competence, (ii) parameter-space protection, however placed, cannot, and (iii) the ordering should transfer across backbones because the locus does. The next three sections test each.

5. The Anatomy Predicts the Remedy Ordering

Table 1 reports the six classical strategies on four backbones and three scenarios, three seeds each; 68 of 72 cells are complete, and the 4 missing cells are marked rather than imputed.[1]

Replay re-grounds the head and reaches the oracle on DIL, on every backbone. ER and DER++ match the joint bound on all four backbones (87.9/88.0 vs. 88.7 on LayoutLMv3, 87.8/87.6 vs. 87.5 on BROS, 77.1/77.4 vs. 77.2 on BERT, 85.7/84.4 vs. 83.8 on LiLT[2]) with BWT of −1 to −3.5. The independent per-dataset bound (88.2/85.3/88.8/77.3) is within a point of joint and of the replay cells, so the question for a continual learner is whether one model captures it at a storage cost below 3× model weights. On Mixed, replay closes most of the gap (60.3–63.0 vs. oracles 65–68 on layout-aware backbones; BERT 50.2/50.7 vs. 55.0); the class-incremental transitions inside the sequence are where residual forgetting sits.

Parameter-space protection does not. EWC and LwF remain at the naive floor on DIL for LayoutLMv3 (41.2, 43.7 vs. 41.3). Both mechanisms act on parameters or on whole-network outputs; neither places an old-task gradient on the head, and the anatomy says that is the only place it would matter. EWC's Fisher penalty, computed at the head-dominant locus, does reduce backward transfer (−2.8 vs. −73 for naive on LayoutLMv3) but at the cost of not fitting the new tasks, the same under-fitting mode as the uniform consolidation probe of §4.

[1] The missing cells (LwF on LiLT; EWC on LiLT/CIL-CORD) exceeded the memory of the 6 GB GPU that ran the secondary-backbone grid, even with mixed precision; they are reported as missing rather than imputed.

[2] Replay slightly exceeds the pooled-data oracle on LiLT; joint on LiLT appears under-fitted by the shared early-stopping rule.



Figure 2. **Forgetting has an address, and protecting it moves it.** Left: linear CKA between checkpoints before and after the second task falls with depth on all four backbones (1.0 = no drift). Right: freezing the head and late layers leaves backward transfer unchanged, (a) whatever the number of frozen layers and (b) on three DIL task orders and a ten-session CIL-CORD stream (LayoutLMv3, seed 42, no memory). The drift relocates to the layers left plastic.

Table 1. Generality grid: average accuracy (mean ± s.d., three seeds) of the six classical strategies on four backbones and three scenarios. *Indep.* is the per-dataset upper bound (one model trained independently on each dataset, three-seed macro over FUNSD/SROIE/CORD). *Joint* is the pooled-data oracle (one model, all three datasets). Independent is only meaningful on DIL, where the task stream is three distinct datasets; CIL-CORD is one dataset split by label and Mixed is a composed stream, so Indep. is undefined there and marked N/A. Grey cells did not fit the available GPU memory (see text). Replay reaches the joint bound on DIL on every backbone; regularisation and distillation stay near the naive floor; on CIL-CORD every method, replay included, sits near the floor.

Scenario	Backbone	Naive	Indep.	Joint	EWC	LwF	ER	DER++
DIL	LayoutLMv3	41.3±1.5	88.2±1.0	88.7±1.1	41.2±1.1	43.7±1.1	87.9±1.4	88.0±0.9
	LiLT	40.1±0.5	85.3±0.8	83.8±1.2	48.4±5.9	–	85.7±0.5	84.4±0.3
	BROS	40.3±1.8	88.8±0.1	87.5±1.3	45.0±4.9	35.2±3.1	87.8±0.4	87.6±1.1
	BERT	39.4±0.3	77.3±0.5	77.2±0.4	46.6±3.0	40.5±0.1	77.1±1.0	77.4±0.6
CIL-CORD	LayoutLMv3	19.0±0.2	N/A	33.3±0.3	0.5±0.6	18.8±0.2	16.0±1.4	18.2±0.3
	LiLT	19.1±0.3	N/A	32.4±0.4	–	–	17.9±0.5	18.0±1.0
	BROS	19.2±0.1	N/A	32.9±0.0	17.3±0.8	23.1±1.2	18.3±1.5	17.7±1.1
	BERT	19.3±0.1	N/A	32.4±0.2	17.9±0.3	24.3±0.3	17.1±1.6	19.2±1.9
Mixed	LayoutLMv3	33.9±0.3	N/A	67.5±0.8	22.0±9.4	35.2±1.2	60.3±5.3	60.8±2.7
	LiLT	30.7±0.7	N/A	61.8±0.6	28.1±3.0	–	57.0±1.4	55.1±2.9
	BROS	33.9±0.4	N/A	65.3±2.1	37.4±3.3	34.4±0.5	61.1±1.4	63.0±0.4
	BERT	23.4±0.3	N/A	55.0±0.6	28.6±0.9	29.4±4.2	50.2±0.6	50.7±1.3

A remedy's benefit is backbone-dependent; the phenomenon is not. Where the naive-to-joint gap agrees to 2.5 points across backbones on Mixed, EWC's benefit over naive spans a 17-point range with a sign flip: −11.8 on LayoutLMv3, −2.6 on LiLT, +3.5 on BROS, +5.3 on BERT (supplementary Sec. C). LwF likewise helps CIL-CORD on BROS and BERT (+4, +5) and nothing on LayoutLMv3. A single-backbone comparison of buffer-free remedies can therefore land on either side of zero; a multi-backbone protocol is not optional for this class of claims.

CIL-CORD is a floor regime for everyone. With six fine-grained classes per session over a single receipt domain, every method, replay included, sits at 16–24 AA against an oracle of 33, and on LayoutLMv3 EWC collapses to 0.5. Part of this floor is the scoring: an oracle trained on every label scores 33 because each session's gold hides the other sessions' entities, and 91.9 under session-restricted scoring (Tab. 3). The rest is real and survives the fix: under restricted scoring replay rises only to 20–25 and naive stays at 19, because what must be re-grounded is not a fixed head but a head whose background class is redefined at every session, and replayed documents carry the *old* labelling of the new classes. Sec. 8 resolves it at the readout.[3]

6. Falsification: Five Buffer-Free Families

If replay works because it re-exercises the head on old-task targets, the strongest test is to seek the same effect without a buffer. We built or ported five buffer-free families on

[3]This is the setting in which the continual-NER literature's distillation/pseudo-labelling protocols are designed to operate; we deliberately do not engineer the O-shift away.

DIL/LayoutLMv3 (naive floor 41.3, replay ≈88) and root-caused each failure per boundary (supplementary Sec. C): TIES/Fisher head merging is a near no-op (21.6 vs. 21.9 unmerged); merging with Gaussian classifier realignment (LCA [35]) halves BWT but kills acquisition (43.9±2.5); subspace transfer (HGT, CUBER) leaves the head unconstrained (39.8 at its own 38.7 floor); isolated head slots cannot re-ground a shared head (42.2); and feature-Gaussian replay (LexMem) recovers only dense-label tasks (66.0, falling to 13.0 mid-task once relational summaries break document consistency). None is a contribution; each is a witness.

One mechanism, and cheap corrections are nulls. The head's output marginal on old tasks snaps to the just-trained task's gold marginal (cosine 0.997–0.998 for naive and LwF against 0.60 to the old task's own; replay stays at 0.99–1.00). Classes never re-exercised (KEY, HEADER) are extinguished to exactly 0.0 F1 under every modality mask. Forgetting is a *readout-marginal snap on a class-asymmetric substrate*, not representation erasure, which is why the feature-Gaussian family [8, 29, 52] recovers only partially (Sec. 7). If the snap were a mis-calibrated prior, correction would reverse it; five kill-tests registered with explicit bars before any run (marginal matching, prior-ratio correction, per-document EM, a KL anchor to the prior-task mixture, balanced-softmax logit adjustment) recover at best −0.1 AA and leave the extinguished classes at 0.0, because old-class mass is $\sim 10^{-6}$, orders below any branch threshold. With the migration result of Sec. 4, buffer-free fixes are ruled out and the question becomes what minimal object a buffer must store.

7. CoLaR: What a Document Memory Must Hold

Any admissible memory must re-exercise the head on inputs that keep their old joint labelling. To find the minimum stored object we run a falsification ladder over *latent* replay: activations banked at a frozen encoder boundary (layer k) and replayed through the plastic upper layers and head, so the stored content varies while the mechanism is fixed (DIL/LayoutLMv3; single-seed diagnostics except where noted; supplementary Sec. C).

Whole documents retain; every summary collapses. Replaying the real banked activations of 50 whole documents per task retains AA 87.3 (k=4). Summaries of the *same* store collapse: a rank-16 spectral summary 41.9, per-class Gaussians 39.4, even real k-means centroids 36.7–39.7, so the failure is not synthesis fidelity. The single-variable control isolates the ingredient: four whole replayed documents retain 63.8 at a matched five-epoch budget, while the identical features on decoupled layout carriers retain 36.7. Whole-document binding is worth +27 AA by itself.

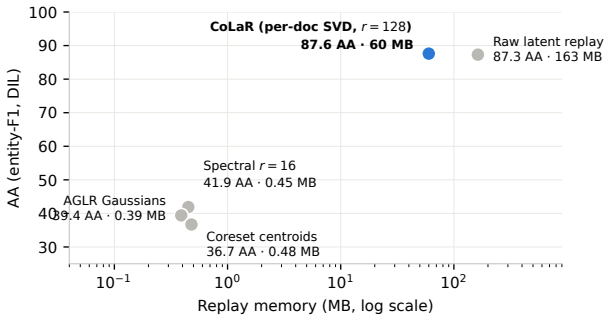


Figure 3. Replay memory versus retention (DIL, log-scale memory). Per-document SVD replay (CoLaR) matches raw latent replay at 2.7× less storage because it preserves whole-document binding; every aggressive summary at ∼0.4 MB breaks that binding and collapses to the naive level.

The ingredient is domain-discriminating. On ViT-B/16 image CIL (CIFAR-100 and ImageNet-R, ten sessions, three seeds; supplementary Sec. C) the ranking inverts: the buffer-free Gaussian-head method SLCA [47] reaches 87.5/65.0 (joint 89.5/77.6), above raw replay (ER 76.2/54.8), above LCA (77.2) and above an uncompressed latent bank of 500 images per class (80.0/55.1), and collapses to 38.0/25.8 without its classifier alignment. The image trunk is undamaged: a ridge readout refit on the trained latent-bank trunk reaches 88.3/72.3, above frozen-trunk RanPAC [26] (85.3/58.5) and above SLCA, so only the readout statistics are stale, and the headroom is coverage of images the bank does not hold rather than the head's form. CoLaR's per-image SVD is lossless against the bank it compresses (79.5 vs. 80.0; 55.3 vs. 55.1). On documents the same idea is null in every form (feature Gaussians as a store 39.4; Gaussian realignment after merging 43.9; Gaussian head alignment on CoLaR 86.2 vs. 86.4) while whole-document replay matches joint. An image carries one label, so a class-conditional Gaussian summarises its readout without loss; a document carries one label per token bound to a position and layout, and no per-class statistic of pooled features keeps that binding.

CoLaR, the proposed memory. The two negatives above (every summary of a store collapses while whole documents retain) say the admissible compressor must keep each document's binding. Individual documents are low-rank even though the pooled feature space is not: one document's tokens share one layout and one vocabulary, so its layer-k activation matrix $H \in \mathbb{R}^{N\times d}$ has fast singular-value decay. We therefore **truncate the SVD per document**: $H_j \approx U_r \Sigma_r V_r^\top$, stored with the document's bounding boxes, attention mask and BIO labels. Replay rebuilds $\hat{H}_j = U_r \Sigma_r V_r^\top$ and injects it at layer k through a forward

Table 2. **Fixed comparison set on LayoutLMv3** (AA, mean ± s.d. over seeds 42/7/123): the two bounds and the ten published baselines (document-side paired slots CPFD and IS3), then our rows; the image-side table is in the supplement. CoLaR, CoLaR-R and the baselines use 10 training epochs per task with early stopping; LexSlot is the 200-exemplar slot-memory recipe. *did not converge on these sessions, which naive (19.0) and joint (33.3) do learn (supplement, "Null cells among the fixed baselines"). "–": not applicable; IS3 on DIL is a single seed (fixed-label port limitation, supplement).

Method	CIL-CORD	DIL	Mixed
Naive (lower bound)	18.98 ± 0.17	41.31 ± 1.54	33.85 ± 0.33
LwF	18.82 ± 0.15	43.66 ± 1.05	35.23 ± 1.17
ER	15.98 ± 1.36	87.85 ± 1.42	60.32 ± 5.25
DER++	18.15 ± 0.30	**88.02 ± 0.88**	**60.82 ± 2.74**
CODA-Prompt	null*	32.51 ± 1.23	8.11 ± 1.52
CL-LoRA (2025)	15.17 ± 3.14	41.14 ± 0.28	31.37 ± 0.63
CPFD (2023)	null*	6.15 ± 7.29	null*
IS3 (2024)	null*	38.29 (1 seed)	30.10 ± 1.18
LCA (2026)	8.15 ± 2.59	43.87 ± 2.45	19.01 ± 3.68
Text-only comparator (no layout or image):			
BERT-base	19.26 ± 0.11	39.42 ± 0.26	23.37 ± 0.27
CoLaR (ours)	20.38 ± 1.34	87.75 ± 0.48	58.75 ± 1.10
CoLaR-R, sliced (ours)	25.38 ± 2.38	–	61.30 ± 1.91
LexSlot (ours)	19.28 ± 0.12	87.33 ± 1.03	35.49 ± 1.39
Joint (upper bound)	33.32 ± 0.33	88.73 ± 1.13	67.51 ± 0.82
Independent (one model per task, task-ID oracle)	–	88.79 ± 0.55	–

hook; the loss is CE on the current batch plus CE on the replayed documents. The memory never stores a raw image or token id. Architecture in Fig. 1; the per-document rank argument is illustrated in the supplement (Sec. C).

CoLaR at r=128 matches raw-exemplar replay. CoLaR retains AA 87.6 (87.8 ± 0.5 over three seeds) at 60 MB, within 1.1 points of the joint oracle, matching raw-exemplar ER (87.9, 200 documents) at a budget that stores no private bytes, and 2.7× smaller than raw latent replay at equal document count (87.3 at 163 MB); r=64 gives 80.6 at 32 MB (Fig. 3). Attempts to shift the memory-side frontier obey a conservation law: class-balanced losses 87.1, coverage-based selection 86.8, entity-weighted compression 87.1 all trade FUNSD for SROIE about 1:1 without moving AA; a three-seed check finds no lever beats the base (balanced 87.8, k-centre 87.0). Read-side extensions (kNN readout blend, metaplastic consolidation) are clean negatives.

Class-incremental documents: live labels help, but the shared softmax is the ceiling. On CIL-CORD under legacy scoring every replay method sits at the naive floor (Tab. 2): CoLaR 20.4 ± 1.3, ER 16.0, DER++ 18.2 against naive 19.0 and joint 33.3. A document banked at session t carries "O" on every entity introduced after t, so replaying it re-teaches the head to suppress exactly the classes it should keep. Refreshing the stored labels from a per-document ledger (the true-label ceiling) gives 26.3 ± 3.6 and a sliced replay loss over the classes alive at banking time gives 25.4 ± 2.4, each above plain CoLaR on every seed (three seeds, pre-registered); teacher-free pseudo-labelling is a registered null (20.0 ± 4.1). Under session-restricted scoring the ordering is unchanged and the gap to the oracle becomes visible: CoLaR 30.4, sliced 42.0, true-label refresh 61.2 (seed 42), joint 91.9. Even with perfect labels the memory recovers a third of the gap, because current documents still label old entities "O" and replayed documents label new ones "O", and both move the same background logit on the same surface forms. The continual-NER comparators built for this failure mode do not converge here (IS3) or collapse (CPFD; Tabs. 2 and 3).

Two registered side results. A drift-free OCR-signature sampler that chooses which stored document to replay (Fig. 1) looked promising on one seed (+1.8 over uniform) and is a three-seed null (+1.8/ + 0.4/ − 0.6); the headline does not depend on it. Replaying pseudo-labelled *public* documents through the same mechanism (zero private bytes) recovers dense-label tasks (FUNSD 76.1) but not one the proxy pool does not cover (SROIE ∼7), a coverage-limited 58.9 (supplementary Sec. B).

8. CoLaR-SBR: Un-sharing the Background

Session-blocked readout. The diagnosis above says the contradiction lives in one logit. We therefore un-share it. For each session b that introduces tags the head gains a block $[\mathrm{O}_b, \mathrm{tags}_b]$ with its own background logit, trained with a block-local softmax. A token labelled "O" in session-t data is *truly* O_t, so every session's annotation is complete within its own block. Supervision follows one rule per (document, block) pair: a document trains block b *fully* when its session annotates every tag of b; as *all-background* when b belongs to a different dataset schema, so none of its entities can occur; and not at all otherwise, because its "O" labels are stale for that block. A session that adds no tags (every DIL step, the Mixed revisit) creates no block, so on DIL the method reduces to CoLaR with one block. The trunk is shared, trained by every block, and kept stable by CoLaR's whole-document latent replay; banked documents supervise the blocks of their own session. Inference needs no task id: per block $e_b = 1 - p_b(\mathrm{O}_b)$, the block with the largest e_b claims the token if $e_b > 0.5$, else the token is "O". Closest prior art is ICE [21] (per-session classifiers on a frozen PLM) and SpanKL [51] (per-type sigmoid heads, no replay); SBR adds a trained trunk and whole-document replay, and every block sees true negatives because each session's data is the full receipt population.

Result. Tab. 3 reports the frozen comparison set under session-restricted scoring. CoLaR-SBR reaches 88.0 ± 1.9 AA on CIL-CORD against joint 91.9 ± 1.3 and the best base-

line, ER, at 24.3 ± 2.4; on Mixed it reaches 89.7 ± 0.1 (two seeds) against joint 87.7 ± 1.1 and ER 74.5. On DIL, where it collapses to CoLaR with one block, it scores 86.2 ± 1.3. Two ablations locate the effect: the same blocks without the bank fall to 20.4 and on a frozen trunk to 14.8, so readout and memory are both necessary and neither sufficient. Tuning the threshold and a per-block bias even on the test set moves AA by at most $+0.6$, so the residual loss is retention, not readout. Storage is the 96 MB bank plus one logit per block. Stacking LexSlot's lexically routed head slots on the blocks moves the three-seed mean by $+1.7$ (CIL-CORD), -0.1 (Mixed) and $+0.8$ (DIL) and halves the seed variance: the two mechanisms partition the same readout.

SBR-Recall: a memory of brain states. The bank is write-only at test time. SBR-Recall also stores, per session, a LoRA [13] delta of the upper trunk (layers $\geq k$; rank 64 on the attention projections and feed-forward layers) and evaluates block b *under the delta of the session that created it*, so e_b is computed by the network state that learned those tags. The base trunk is frozen after the first session, so old sessions are forgetting-free by construction and a delta never goes stale, at one upper-trunk forward per distinct delta at inference. Routing stays task-agnostic and per token, which separates this from parameter isolation with a task oracle and from CL-LoRA's shared softmax over per-task adapters (18.6 here despite a diagonal of 93). On CIL-CORD SBR-Recall reaches 92.8 ± 0.5 AA over three seeds (92.7/93.4/92.4, BWT $+0.6/+0.4/-0.1$), above the joint oracle on every seed with every retention column flat to within a point, and needs no bank (54 MB of fp16 deltas, 7.1M trainable parameters). At rank 8 on the attention projections alone it scores 80.5 with exact retention but two sessions plateauing near 65: the isolation is sound and the delta's capacity is what CORD's look-alike price classes demand. On the mixed stream the picture inverts: without a bank it scores 44.7. Across schemas every current document adds an all-background term for the other sessions' blocks, and with no positive term to balance it those blocks learn to abstain until their own domain is revisited. Restoring the 50-document bank supplies the counterweight and gives 85.9, below CoLaR-SBR's 89.7: the frozen base removes the need for replay only on a single-schema stream, and the shared-trunk SBR row stands as the Mixed result.

9. Limitations

Scope of each claim. Finding 1 rests on four backbones, three seeds and two instruments; cross-backbone *magnitude* comparisons are underpowered and not claimed. The benchmark ordering is three seeds on 68 of 72 cells, missing cells disclosed. Finding 2 and the buffer-content ladder are seed-42 DIL/LayoutLMv3 diagnostics with three-seed checks only where stated; CoLaR's headline is three-seed.

Table 3. **Session-restricted scoring on LayoutLMv3** (AA, mean $\pm$ s.d.; superscript = number of seeds; unmarked baseline rows are seeds 42/7 with the third seed in progress). Same frozen reference set as Tab. 2, same runs, re-scored with each session's predictions restricted to its label set. Baseline rows use bf16 autocast, SBR rows fp32. [a] one LwF seed per scenario collapsed to all-"O" (Mixed seed 7, CIL-CORD seed 123) and is being re-run. [b] with the 50-document bank; without it 44.7, see text.

Method	CIL-CORD	Mixed
Naive (lower bound)	18.9±0.3 [3]	44.0±0.5
LwF	12.3±10.7 [3,a]	23.9±33.8 [a]
ER	24.3±2.4 [3]	74.5±1.6
DER++	20.0±1.8	74.5±0.8
CODA-Prompt	14.9±1.4	11.3±1.1
CL-LoRA (2025)	18.6±0.1	44.8±1.3
CPFD (2023)	1.4±1.9	7.7±10.5
IS3 (2024)	6.6±1.5	20.3±28.7
LCA (2026)	10.3±4.8	38.0 [1]
CoLaR	30.4±2.2	75.6 [1]
CoLaR-R, sliced	42.0±3.1	80.2 [1]
LexSlot	19.0±0.0	46.9±5.6
CoLaR-SBR (ours)	88.0±1.9 [3]	**89.7±0.1**
+ LexSlot head slots	89.7±0.5 [3]	89.6±1.0 [3]
SBR-Recall (ours)	**92.8±0.5** [3]	85.9 [1,b]
Joint (upper bound)	91.9±1.3 [3]	87.7±1.1

Four document cells in three reference rows are disclosed nulls (Tab. 2), none imputed.

What we do not claim. CoLaR is still a replay method with a store proportional to the number of past tasks. Its hyperparameters (k=4, r=128), tuned on LayoutLMv3, transfer near-losslessly to BROS (87.2 vs. ER 87.8 on DIL) but trail raw replay by 5–9 points on text-heavy BERT and LiLT; per-backbone tuning was not attempted, and LiLT's separate layout stream is not stored. Tab. 3 is two-seed for most baseline rows until the queue completes; SBR's Mixed row is two-seed and SBR-Recall's Mixed row single-seed, each claimed only as such, and neither SBR variant has been run on the secondary backbones. Every run early-stops on the current task's evaluation split, a signal all rows share (a held-out-split re-run is in the supplement). The image cells use one house recipe for every row, which trails the published SLCA/RanPAC recipes by a uniform offset, so we claim the ordering, not the level; CODA-Prompt and CL-LoRA run without their official old-class logit masking (masked re-run in the supplement).

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